

Canaries in the Coal Mine? Six Facts about the Recent Employment Effects of Artificial Intelligence

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Abstract

This paper examines changes in the labor market for occupations exposed to generative artificial intelligence using high-frequency administrative data from the largest payroll software provider in the United States. We present six facts that characterize these shifts. We find that since the widespread adoption of generative AI, early-career workers (ages 22-25) in the most AI-exposed occupations have experienced a 13 percent relative decline in employment even after controlling for firm-level shocks. In contrast, employment for workers in less exposed fields and more experienced workers in the same occupations has remained stable or continued to grow. We also find that adjustments occur primarily through employment rather than compensation. Furthermore, employment declines are concentrated in occupations where AI is more likely to *automate*, rather than *augment*, human labor. Our results are robust to alternative explanations, such as excluding technology-related firms and excluding occupations amenable to remote work. These six facts provide early, large-scale evidence consistent with the hypothesis that the AI revolution is beginning to have a significant and disproportionate impact on entry-level workers in the American labor market.

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firm regardless of AI exposure. For workers aged 22-25, we find a 12 log-point decline in relative employment for the most AI-exposed quintiles compared to the least exposed quintile, a large and statistically significant effect. Estimates for other age groups are much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

Fifth, the labor market adjustments are visible in employment more than compensation. In contrast to our findings for employment, we find little difference in annual salary trends by age or exposure quintile, suggesting possible wage stickiness. If so, AI may have larger effects on employment than on wages, at least initially.

Sixth, the above facts are largely consistent across various alternative sample constructions. We find that our results are not driven solely by computer occupations or by occupations susceptible to remote work and outsourcing. We also find that the AI exposure taxonomy did not meaningfully predict employment outcomes for young workers further back in time, before the widespread use of LLMs, including during the unemployment spike driven by the COVID-19 pandemic. The patterns we observe in the data appear most acutely starting in late 2022, around the time of rapid proliferation of generative AI tools.⁴ They also hold for both occupations with a high share of college graduates and ones with a low college share, suggesting deteriorating education outcomes during COVID-19 do not drive our results. For non-college workers, we find evidence that experience may serve as less of a buffer to labor market disruption, as low college share occupations exhibit divergent employment outcomes by AI exposure up to age 40.

While we caution that the facts we document may in part be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment. We intend to continue to track the data on an ongoing basis to assess whether these trends change in the future.

Why might AI adversely affect exposed entry-level workers more than other age groups? One possibility is that, by nature of the model training process, AI replaces codified knowledge, the “book-learning” that forms the core of formal education. AI may be less capable of replacing

⁴OpenAI introduced ChatGPT in November 2022.

f indexes firms, q indexes Eloundou et al. (2024) exposure quintiles, and t indexes months, with $t = -1$ corresponding to October 2022. The outcome variable $y_{f,q,t}$ is employment in f, q, t . Equation 4.1 is a Poisson event study regression controlling for firm-quintile effects, $\alpha_{f,q}$, and firm-time effects, $\beta_{f,t}$. The firm-time effects absorb aggregate firm shocks that impact each exposure quintile equally. The firm-quintile effects adjust for baseline differences in hiring across quintiles within the firm. The coefficients of interest, $\gamma_{q,t}$, measure differential changes in employment growth across quintiles after accounting for firm-time effects and firm-quintile effects.¹⁷

We run this regression separately for each age group. For each regression, we restrict to firms that hire at least 10 workers within the age group in every period of the sample. Further, $\sum_t y_{f,q,t}$ must equal at least 100 for each q , meaning that the firm must at least employ on average about 2 workers from each exposure quintile across months in the sample.¹⁸ Standard errors are clustered by firm.

Results are in Figure 9, which plots the $\gamma_{q,t}$ coefficients for each age group. For workers aged 22-25, estimates for higher quintiles are large and statistically significant, with a 12 log point decline in relative employment comparable in magnitude to the estimates in the raw data in Figure 3. Estimates for other age groups are generally much smaller in magnitude and not statistically significant. These findings imply that the employment trends we observe are not driven by differential shocks to firms that employ a disproportionate share of AI-exposed young workers.

One alternative confounder that would not be controlled for with firm-time effects is that even conditional on the firm workers with high AI exposure were excessively hired after the COVID-19 pandemic, leading to a subsequent contraction in their hiring. To assess such alternatives we consider various other robustness checks in Section 4.6, such as removing computer occupations and conditioning on whether the occupation is amenable to work from home.

¹⁷Because of zero counts in the outcome variable, we estimate a Poisson regression instead of an OLS regression in logs following guidance from Chen and Roth (2024).

¹⁸Results are not sensitive to these restrictions, though there must be at least one non-zero value in each firm-month and each firm-quintile for observations to not get dropped in the Poisson regression.

consistent with the case studies above that show employment changes are visible across a range of occupations. Results with firm-time fixed effects in Figure 9 further show that our findings are robust to firm- or industry-level shocks that impact general hiring trends. These results indicate that our findings are not specific to technology roles.

Remote Work Figures A6 and A7 show results for occupations amenable to remote work (telework) and those that are not, according to Dingel and Neiman (2020).²² We find that, for young workers, more exposed occupations have slower employment growth, both in teleworkable occupations and in non-teleworkable occupations. The results for non-teleworkable occupations in particular suggest that our findings are not driven by outsourcing or work-from-home disruptions, at least solely.²³

Longer Sample Figure A8 shows results when extending the balanced sample of firms to 2018. This reduces the sample size and makes the data somewhat noisier. Nonetheless, the trends remain largely ordered by exposure in the post-GPT era, whereas this is not the case before 2022. A concern is that for the Eloundou et al. (2024) measures the most exposed quintile had slower employment growth starting around 2020. This is not the case for the Anthropic exposure measures, shown in Figures A9, A10, and A11. For these measures the most exposed groups have comparable employment growth throughout the period before generative AI, with divergent trends afterwards.

Changes in Education Another possibility is that the changes we observe are influenced by worsening education outcomes during the COVID-19 Pandemic. COVID-19 led to persistent harms in education outcomes (Kuhfeld and Lewis, 2025). As more educated workers on average have greater measured AI exposure, deterioration in the quality of education in recent years could influence the trends we observe.²⁴ In Figure A12 we show trends for occupations in which greater than

²²Whether an occupation is amenable to remote work is positively correlated with AI exposure. Only two teleworkable occupations fall in the lowest quintile of estimated AI exposure according to the GPT-4 β measure. Likewise few non-teleworkable occupations fall in the highest quintile of AI exposure. For this reason in Figure A6 we pool together the two lowest AI exposure quintiles into one group. In Figure A7 we pool together the two highest AI exposure quintiles.

²³Non-teleworkable occupations with high AI exposure include bank tellers, travel agents, and tax preparers.

²⁴Chandar (2025a) finds that declines in average skill levels for college graduates explain a sizable share of the slowdown in the growth of the college wage gap in recent decades.

- Fifth, these labor market adjustments are more visible in employment than in compensation.
- Sixth, we find that these patterns hold in occupations unaffected by remote work and across various alternative sample constructions.

While our main estimates may be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to significantly affect entry-level employment.

The adoption of new technologies typically leads to heterogeneous effects across workers, resulting in an adjustment period as workers reallocate from displaced forms of work to new forms with growing labor demand (Autor et al., 2024). Such endogenous adjustment may already be happening with AI, with emerging evidence of shifts in college majors away from AI-exposed categories such as computer science (Horowitz, 2025). Past transitions such as the IT revolution ultimately led to robust growth in employment and real wages following physical and human capital adjustments, with some workers benefiting more than others (Bresnahan et al., 2002; Brynjolfsson et al., 2021).

Tracking employment trends on an ongoing basis will help determine if the adjustment to AI follows a similar pattern. Consequently, we will continue to monitor these outcomes to assess whether the trends documented in the paper accelerate in the future. Future work would benefit from better firm-level AI adoption data, which would provide sharper variation for estimating plausible causal effects of AI on employment.

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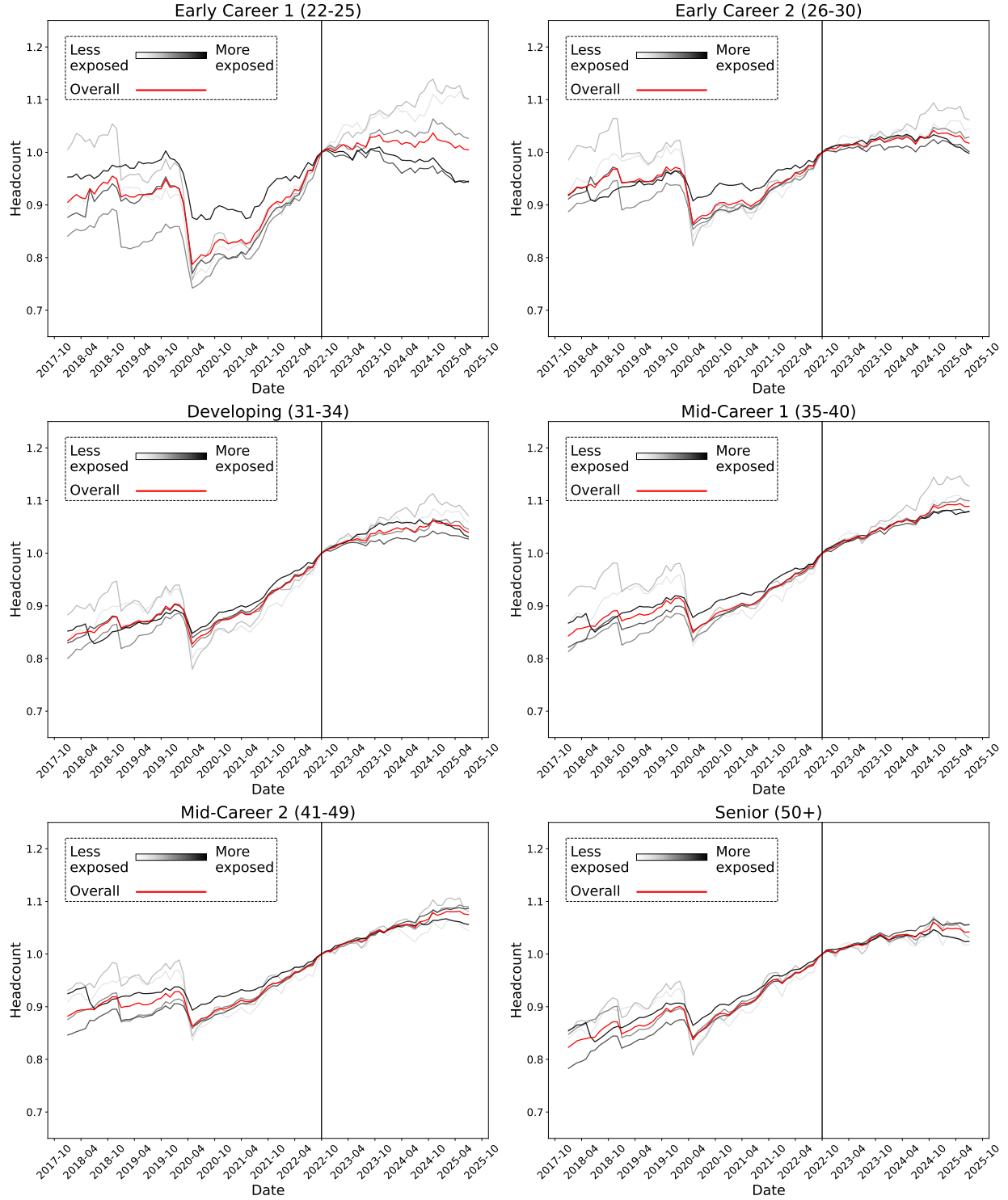


Figure A8: Employment changes by age and exposure quintile using measures from [Eloundou et al. \(2024\)](#). Data is from 2018 to 2025.